Calc II Section Notes

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Chapter 1

Integrals

Week 1 (8/31-9/3)

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Problem 1.1 (discuss with your teammates)**.** You are an open water diver, which of the following sharks would you be excited to see? Rank from 1 to 5 most excited to least excited.

Shark Type	Diet	Humans	Appearance
Tiger shark	Sea turtles, rays, other sharks, fish, seabirds, dolphins, squid, crustaceans, and carrion	142 (39 fatal)	Long body with a broad head and dark vertical bands along the sides.
Hammerhead shark	Fish, squid, crustaceans, rays, and smaller sharks	18 (0 fatal)	Distinctive hammer-shaped head with eyes positioned far apart.
Nurse shark	Fish, stingrays, octopus, squid, clams, and crustaceans	9 (0 fatal)	Rounded head with small eyes and two barbels near the mouth.
Thresher shark	Mostly schooling fish such as herring and mackerel; also squid	0 listed	Extremely long upper tail fin, nearly as long as the rest of the body.
Great white shark	Fish, rays, other sharks, squid, and marine mammals	351 (59 fatal)	Muscular body with prominent triangular teeth. Powerful and heavy-looking.

Problem 1.2. Approximate the area under $f(x) = x^2$ on $[0, 1]$ using Riemann sums with $N = 2, 3$ and some choice of sample points. Then think about the limit as $N \to \infty$. Hint:

$$\sum_{i=1}^{N} i = 1 + 2 + \cdots + N = \frac{N(N+1)}{2}, \quad \sum_{i=1}^{N} i^2 = 1 + 4 + 9 + \cdots + N^2 = \frac{N(N+1)(2N+1)}{6}$$

Proof. We will use left endpoints. When $N = 2$, $x_1 = 0, x_2 = 1/2$. Thus

$$R_2 = 0 + \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$$

When $N = 3$, $x_1 = 0, x_2 = 1/3, x_3 = 2/3$. Then

$$R_3 = 0 + \frac{1}{9} \cdot \frac{1}{3} + \frac{4}{9} \cdot \frac{1}{3} = \frac{5}{27}$$

Now we compute when $N \to \infty$

$$R = \lim_{N\to\infty} \sum_{i=0}^{N-1} \left(\frac{i}{N}\right)^2 \frac{1}{N} = \frac{1}{6}$$

□

Problem 1.3. Compute and interpret the limit

$$\lim_{N\to\infty} \sum_{i=1}^{N} \left(\frac{i}{N^2} - \frac{i^2}{N^3}\right)$$

Proof. We first compute it:

$$\sum_{i=1}^{N} \left(\frac{i}{N^2} - \frac{i^2}{N^3}\right) = \frac{N(N+1)}{2N^2} - \frac{N(N+1)(2N+1)}{6N^3} \tag{1.1}$$

Letting $N \to \infty$,

$$\lim_{N\to\infty} \sum_{i=1}^{N} \left(\frac{i}{N^2} - \frac{i^2}{N^3}\right) = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

Now to interpret it, we can write it as a Riemann sum:

$$\sum_{i=1}^{N} \left(\frac{i}{N^2} - \frac{i^2}{N^3}\right) = \sum_{i=1}^{N} \frac{1}{N} \left(\frac{i}{N} - \frac{i^2}{N^2}\right)$$

Then $\Delta x = \frac{1}{N}$, and $x_i = \frac{i}{N}$. Letting $N \to \infty$, this is the Riemann sum of the following

$$\int_0^1 (x - x^2)\, dx$$

□

Problem 1.4. Compute

$$\int_{-1}^{5} (1 + 3x)\, dx$$

using Riemann sums.

Proof. We divide the interval $[-1, 5]$ into N intervals, then $\Delta x = 6/N$, and if we use right endpoints, the ith sample point is $x_i = -1 + 6i/N$. Then

$$\sum_{i=1}^{N} f(x_i)\Delta x = \sum_{i=1}^{N} (-2 + 18i/N)\frac{6}{N}$$

letting $N \to \infty$, the answer is 42. □

Week 2 (9/7-9/11)

Problem 1.5. Compute the following integral:

$$\int_1^3 \frac{\ln x}{x(1+\ln^2 x)}dx$$

Proof. Let $u = 1 + \ln^2 x$, then $du = \frac{2\ln x}{x}dx$, and

$$\int_1^3 \frac{\ln x}{x(1+\ln^2 x)}dx = \int_1^{1+\ln^2 3} \frac{1}{2u}du = \frac{1}{2}\ln(1+\ln^2 3)$$

□

Problem 1.6. Compute the following integral:

$$\int_0^{1/2} \frac{x}{\sqrt{1-x^4}}dx$$

Proof. Let $u = x^2, du = 2xdx$, then

$$\int_0^{1/2} \frac{x}{\sqrt{1-x^4}}dx = \int_0^{1/4} \frac{1}{2\sqrt{1-u^2}}du = \frac{1}{2}\left(\arcsin 1/4 - \arcsin 0\right)$$

□

Problem 1.7. Compute the following integral:

$$\int_1^2 \frac{e^x}{(1+e^x)\ln(1+e^x)}dx$$

Proof. Let $u = \ln(1+e^x)$, then $du = \frac{e^x}{1+e^x}dx$, thus

$$\int_1^2 \frac{e^x}{(1+e^x)\ln(1+e^x)}dx = \int_{\ln(1+e)}^{\ln(1+e^2)} \frac{1}{u}du = \ln\left(\frac{\ln(1+e^2)}{\ln(1+e)}\right)$$

□

Chapter 2

Applications of Integrals

Week 3 (9/14-9/18)

Problem 2.1. Compute the area enclosed by the following curves:

1. $$y = \frac{1}{x}, y = \frac{1}{x^2}, x = 2$$
2. $$x = 1 - y^2, x = y^2 - 1$$
3. $$x = \frac{\pi}{2}, x = \pi, y = \sin x, y = x$$
4. $$y = \sqrt[3]{2x}, y = \frac{1}{8}x^2, x = 0, x = 6$$

Proof. 1.
$$\int_1^2 \left(\frac{1}{x} - \frac{1}{x^2}\right) dx$$

2.
$$\int_{-1}^1 (1 - y^2) - (y^2 - 1) dy$$

3.
$$\int_{\pi/2}^{\pi} (x - \sin x)\, dx$$

4.
$$\int_0^4 \left(\sqrt[3]{2x} - x^2/8\right) dx + \int_4^6 \left(x^2/8 - \sqrt[3]{2x}\right) dx$$

□

Problem 2.2. Write down the integrals for the following solids of revolution using the washer method. DO NOT SOLVE.

1. $y = x, y = x^3, x \geq 0$, rotated about the x-axis.
2. $y^2 = x, x^2 = y$, rotated about $y = 1$.
3. $y^2 = x, x^2 = y$, rotated about the y-axis.
4. $x = 0, y = x^2, y = -2x + 3$ around the y-axis.

Proof.

1. The cross-sectional area in the x-direction is $\pi(x^2 - (x^3)^2)$, thus
$$S = \int_0^1 \pi(x^2 - x^6)dx$$
2. The cross-sectional area along the x-axis is
$$A = \pi\left[(1 - x^2)^2 - (1 - \sqrt{x})^2\right]$$
Thus the solid volume is
$$S = \int_0^1 \pi\left[(1 - x^2)^2 - (1 - \sqrt{x})^2\right] dx$$
3. The cross-sectional area along the y-axis is
$$\pi\left[(\sqrt{y})^2 - (y^2)^2\right] dy$$
Thus the solid volume is
$$S = \int_0^1 \pi\left[(\sqrt{y})^2 - (y^2)^2\right] dy$$
4. This needs to be separated into two terms:
$$S = \int_0^1 \pi(\sqrt{y})^2 dy + \int_1^3 \pi\left(\frac{(3-y)}{2}\right)^2 dy$$

□

Chapter 3

Techniques of Integration

Week 4 (9/21-9/25)

Problem 3.1. Compute

$$\int (\ln x)^2 dx$$

Proof. We do integration by parts twice:

$$\begin{cases} u = (\ln x)^2, dv = dx \\ du = \frac{2\ln x}{x} dx, v = x \end{cases}$$

Thus

$$\begin{aligned} \int (\ln x)^2 dx &= x(\ln x)^2 - \int 2\ln x dx \\ &= x(\ln x)^2 - 2\left(x\ln x - \int 1 dx\right) \\ &= x(\ln x)^2 - 2x\ln x + 2x + C \end{aligned}$$

□

Problem 3.2. Recall

$$\int e^x \sin x dx = \frac{1}{2} e^x (\sin x - \cos x) + C$$

Compute

$$\int e^{2\theta} \sin(3\theta) d\theta$$

Proof. Let $f' = e^x, g = \sin x$, then

$$\begin{aligned} \int e^x \sin x dx &= e^x \sin x - \int e^x \cos x dx \\ &= -\left(e^x \cos x + \int e^x \sin x\right) \qquad \text{(IBP twice)} \end{aligned}$$

This gives the implies result. Equivalently, we let $u = \sin x, du = \cos x dx, dv = e^x dx, v = e^x$. Now we compute

$$\begin{aligned}\int e^{2\theta}\sin(3\theta)d\theta &= \frac{1}{2}e^{2\theta}\sin(3\theta) - \frac{3}{2}\int e^{2\theta}\cos(3\theta)d\theta \\ &= \frac{1}{2}e^{2\theta}\sin(3\theta) - \frac{3}{2}\left(\frac{1}{2}e^{2\theta}\cos(3\theta) + \frac{3}{2}\int e^{2\theta}\sin(3\theta)d\theta\right)\end{aligned}$$

which implies

$$\int e^{2\theta}\sin 3\theta d\theta = \frac{4}{13}\left(\frac{1}{2}e^{2\theta}\sin 3\theta - \frac{3}{4}e^{2\theta}\cos 3\theta\right) + C$$

□

Problem 3.3. Compute

$$\int_0^1 (x^2+1)e^{-x}dx$$

Proof.

$$\begin{aligned}\int_0^1 (x^2+1)e^{-x}dx &= -e^{-x}(x^2+1)\Big|_0^1 + \int_0^1 e^{-x}2x dx \\ &= (1-2e^{-1}) - e^{-x}2x\Big|_0^1 + \int_0^1 e^{-x}2dx \\ &= 1 - 4e^{-1} - 2e^{-x}\Big|_0^1 \\ &= 3 - 6e^{-1}\end{aligned}$$

□

Week 4 (9/28-10/2)

Section 7.2 and 7.3

Proposition 3.1. For $\sin, \cos$:

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x) \qquad \text{and} \qquad \cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$2 \sin x \cos x = \sin 2x$$

For example, $\int \sin^4 x$.

Proposition 3.2. For $\tan, \sec$:

$$\tan^2 + 1 = \sec^2 x \qquad \text{and} \qquad \sec^2 x = \tan^2 x + 1$$

$$\int \tan x \, dx = \ln|\sec x| + C \qquad \text{and} \qquad \int \sec x \, dx = \ln|\sec x + \tan x| + C$$

For example, $\int \tan^3 x$.

Proposition 3.3.

(a) $\sin A \cos B = \frac{1}{2}[\sin(A - B) + \sin(A + B)]$

(b) $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$

(c) $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$

Problem 3.4. $\int \tan^2 x \, dx$

Proof.

$$\begin{aligned} \int \tan^2 x \, dx &= \int (\sec^2 x - 1) dx \\ &= \tan x - x + C \end{aligned}$$

□

Problem 3.5. $\int x \sin^2(x^2) \, dx$

Proof. First let $u = x^2$, then

$$\begin{aligned} \int x \sin^2(x^2) \, dx &= \frac{1}{2} \int \sin^2 u du \\ &= \frac{1}{4} \int 1 - \cos 2u du \\ &= \frac{x^2}{4} - \frac{\sin(2x^2)}{8} + C \end{aligned}$$

□

Problem 3.6. $\int \tan^3 x \sec x \, dx$

Proof. The power of $\tan$ is odd, so we want to save a factor of $\tan x \sec x$:

$$\begin{aligned}\int \tan^3 x \sec x \, dx &= \int \tan^2(\tan x \sec x) dx \\ &= \int (1 - \sec^2 x)(\tan x \sec x) dx \\ &= \int (1 - u^2) du\end{aligned}$$

□

Problem 3.7. $\int \sin 3x \sin 6x \, dx$

Proof.

$$\begin{aligned}\int \sin 3x \sin 6x \, dx &= \frac{1}{2}(\cos(3x) - \cos 9x) \\ &= \frac{1}{6}\sin 3x - \frac{1}{18}\sin 9x + C\end{aligned}$$

□

Problem 3.8. $\int \frac{x}{\sqrt{x^2 - 7}} \, dx$

Proof. Let $x = \sqrt{7}\sec\theta$, then

$$\begin{aligned}\int \frac{x}{\sqrt{x^2 - 7}} \, dx &= \int \frac{\sqrt{7}\sec\theta}{\sqrt{7}\tan\theta} \sqrt{7}\sec\theta\tan\theta d\theta \\ &= \int \sqrt{7}\sec^2 d\theta \\ &= \sqrt{7}\tan\theta + C \\ &= \sqrt{x^2 - 7} + C\end{aligned}$$

□

Problem 3.9. $\int_0^1 \sqrt{x^2 + 1} \, dx$

Proof. We let $x = \tan\theta$, then $dx = \sec^2\theta d\theta$, thus

$$\int_0^1 \sqrt{x^2 + 1} \, dx = \int_0^{\frac{\pi}{4}} \sec\theta \cdot \sec^2\theta d\theta$$

then one would use integraion by parts $u = \sec\theta, dv = \sec^2\theta d\theta$, to find

$$\begin{aligned}\int_0^1 \sqrt{x^2+1}\,dx &= \sec\theta\tan\theta - \int \sec\theta\tan^2\theta d\theta \\ &= \sec\theta\tan\theta - \int \sec^3\theta d\theta + \int \sec\theta d\theta\end{aligned}$$

which implies

$$\int_0^1 \sqrt{x^2+1}\,dx = \frac{1}{2}\left(\sec\theta\tan\theta + \ln|\sec\theta+\tan\theta|\right)\Big|_0^{\pi/4}$$

□

Week 5 (10/5-10/9)

Section 7.4 Partial Fractions.

Problem 3.10. Evalute the following integral:

$$\int \frac{1}{\sqrt{x}-\sqrt[3]{x}}dx$$

Hint: let $u = x^{\frac{1}{6}}$.

Proof.

$$\begin{aligned}\int \frac{1}{\sqrt{x}-\sqrt[3]{x}}dx &= \int \frac{6u^5}{u^3-u^2}du \\ &= 6\int \frac{u^3}{u-1}du \\ &= 6\int u^2+u+1+\frac{1}{u-1}du \\ &= 6\left(\frac{u^3}{3}+\frac{u^2}{2}+u+\ln|u-1|\right)+C\end{aligned}$$

then one can substitute $u = x^{\frac{1}{6}}$ in. □

Problem 3.11. Evalute the following integral:

$$\int x\tan^{-1}x dx$$

Proof.

$$\begin{aligned}\int x\tan^{-1}x dx &= \frac{x^2}{2}\tan^{-1}x - \frac{1}{2}\int \frac{x^2}{x^2+1}dx \\ &= \frac{1}{2}x^2\tan^{-1}x - \frac{1}{2}x + \frac{1}{2}\int \frac{1}{x^2+1}dx \\ &= \frac{1}{2}x^2\tan^{-1}x - \frac{1}{2}x + \frac{1}{2}\tan^{-1}x + C\end{aligned}$$

for some constant C. □

Problem 3.12. Evalute the following integral:

$$\int \frac{\sec^2 t}{\tan^2 t+3\tan t+2}dt$$

Proof. Let $u = \tan t$,

$$\begin{aligned}\int \frac{\sec^2 t}{\tan^2 t+3\tan t+2}dt &= \int \frac{1}{u^2+3u+2}du \\ &= \int -\frac{1}{u+2}du + \int \frac{1}{u+1}du \\ &= \ln\left|\frac{u+1}{u+2}\right| + C\end{aligned}$$

for some constant C. □

Ways to solve integrals so far: u-substitution, integration by parts, trig substitution, partial fractions, improper integrals.

Week 6 (10/12-10/16)

Problem 3.13. Compute the following improper integral:

$$\int_1^\infty \frac{\ln x}{x} dx$$

Proof. We compute

$$\lim_{t\to\infty} \int_1^t \frac{\ln x}{x} dx$$

where

$$\int_1^t \frac{\ln x}{x} dx = \int_0^{\ln t} u du = \frac{\ln^2 t}{2} \to \infty \quad \text{as } t \to \infty$$

□

Problem 3.14. Compute the following improper integral:

$$\int_2^\infty y e^{-3y} dy$$

Proof. We are computing

$$\lim_{s\to\infty} \int_2^s y e^{-3y} dy$$

where

$$\begin{aligned}\int_2^s y e^{-3y} dy &= -\frac{y e^{-3y}}{3}\Big|_2^s + \int_2^s \frac{e^{-3y}}{3} dy \\ &= \frac{-(3s+1)e^{-3s}}{9} + \frac{7}{9}e^{-6} \\ &= \frac{7e^{-6}}{9} \quad \text{as } s \to \infty\end{aligned}$$

□

Definition 3.1 (Midpoint Rule).

$$\int_a^b f(x)\,dx \approx M_n = \Delta x \left[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)\right],$$

where

$$\Delta x = \frac{b-a}{n},$$

and

$$\bar{x}_i = \frac{1}{2}(x_{i-1} + x_i) = \text{midpoint of } [x_{i-1}, x_i].$$

Definition 3.2 (Trapezoidal Rule)**.**

$$\int_a^b f(x)\,dx \approx T_n = \frac{\Delta x}{2}\left[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)\right],$$

where

$$\Delta x = \frac{b-a}{n}, \qquad x_i = a + i\Delta x.$$

Definition 3.3 (Simpson's Rule)**.**

$$\int_a^b f(x)\,dx \approx S_n = \frac{\Delta x}{3}\big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\big],$$

where n is even and

$$\Delta x = \frac{b-a}{n}.$$

Proposition 3.4 (Error Bound for Simpson's Rule)**.** Suppose that

$$|f''(x)| \le K \qquad \text{for } a \le x \le b.$$

If E_T and E_M are the errors in the Trapezoidal and Midpoint Rules, then

$$\boxed{|E_T| \le \frac{K(b-a)^3}{12n^2}} \qquad \text{and} \qquad \boxed{|E_M| \le \frac{K(b-a)^3}{24n^2}}.$$

Now suppose that

$$|f^{(4)}(x)| \le C \qquad \text{for } a \le x \le b.$$

If E_S is the error involved in using Simpson's Rule, then

$$\boxed{|E_S| \le \frac{C(b-a)^5}{180n^4}}.$$

Problem 3.15. Consider the following integral

$$\int_0^1 \cos(x^2)dx$$

Given n, one can compute M_n, T_n. Find n large enough such that M_n or T_n have an error with size no more than 0.0001.

Proof. One can take the derivative to find

$$f'(x) = -2x\sin(x^2), \quad f''(x) = -2\sin(x^2) - 4x^2\cos(x^2)$$

thus $|f''(x)| \le 6$. This gives

$$0.0001 \ge \frac{6}{12n^2} \Rightarrow n \ge \sqrt{5000}$$

for the trapezoidal estimation;

$$0.0001 \ge \frac{6}{24n^2} \Rightarrow n \ge 50$$

□

Chapter 4

Infinite Sequences and Series

Definition 4.1 (limit of sequences). Let $\{a_n\}$ be a sequence, then we say a_n converges to a limit L if given any $\varepsilon > 0$, we can find N large enough such that for $n \geq N$, we have

$$|a_n - L| < \varepsilon$$

Problem 4.1. Does the following sequence have a limit? If so, compute it.

$$a_n = \sin\left(\frac{n\pi}{2}\right)$$

Proof. The sequence looks like

$$\{1, 0, -1, 0, 1, 0, \dots\}$$

so the limit doesn't exist. □

Problem 4.2. Does the following sequence have a limit? If so, compute it.

$$\ln(n+1) - \ln(n)$$

Proof. Choose n large enough,

$$\ln\left(\frac{n+1}{n}\right) = \ln\left(\frac{1+\frac{1}{n}}{1}\right) \to 0$$

as $n \to \infty$. □

Problem 4.3. Does the following sequence have a limit? If so, compute it.

$$a_n = \sqrt{n^2+1} - n$$

Proof. We see

$$\begin{aligned}\sqrt{n^2+1} - n &= \frac{(\sqrt{n^2+1} - n)(\sqrt{n^2+1} + n)}{\sqrt{n^2+1} + n} \\ &= \frac{1}{\sqrt{n^2+1} + n} \to 0\end{aligned}$$

as $n \to \infty$. □

Week 8 (10/26-10/30)

Proposition 4.1. If a series $\sum_{n=1}^{\infty} a_n$ converges, then $a_n \to 0$ as $n \to \infty$.



Warning 4.1. When $a_n \to 0$, it doesn't necessarily imply $\sum_{n=1}^{\infty} a_n < \infty$. Counterexample: $\sum_{n=1}^{\infty} \frac{1}{n}$.

Problem 4.4. Let

$$a_n = \frac{2n}{3n+1}.$$

(a) Determine whether $\{a_n\}$ is convergent.

(b) Determine whether $\displaystyle\sum_{n=1}^{\infty} a_n$ is convergent.

Proof. 1. $\{a_n\}$ as a sequence is convergent, and

$$\lim_{n\to\infty} a_n = \frac{2}{3}$$

2. Since $\lim_{n\to\infty} a_n \neq 0$, the series cannot be convergent.

□

Problem 4.5. Determine whether the series is convergent or divergent. If convergent, find its sum.

$$\sum_{n=1}^{\infty} \frac{e^n}{n^2}$$

Proof. The series does not converge.

$$\lim_{n\to\infty} \frac{e^n}{n^2} = \lim_{n\to\infty} \frac{e^n}{2n} = \lim_{n\to\infty} \frac{e^n}{2} = \infty$$

by L'Hospital. □

Problem 4.6. Determine whether the series is convergent or divergent. If convergent, find its sum.

$$\sum_{n=2}^{\infty} \frac{2}{n^2-1}$$

Proof. The series converges. We can apply the comparison test with $\frac{n^2}{2}$:

$$n^2 - 1 \geq \frac{n^2}{2}$$

for all $n \geq 2$. Thus,

$$\sum_{n=2}^{\infty} \frac{2}{n^2-1} \leq \sum_{n=2}^{\infty} \frac{2}{n^2/2} < \infty$$

Now we compute its sum:

$$\sum_{n=2}^{\infty} \frac{2}{n^2-1} = \sum_{n=2}^{\infty} \frac{1}{n-1} - \frac{1}{n+1}$$

The partial sum $s_k = a_1 + \cdots + a_k$ is then

$$s_k = \left(1 - \frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \cdots + = 1 + \frac{1}{2} - \frac{1}{k} - \frac{1}{k+1}$$

Then letting $k \to \infty$, we see the sum is $3/2$. □

Proposition 4.2. Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms.

(i) If $\sum b_n$ is convergent and $a_n \leq b_n$ for all n, then $\sum a_n$ is also convergent.

(ii) If $\sum b_n$ is divergent and $a_n \geq b_n$ for all n, then $\sum a_n$ is also divergent.

Proposition 4.3. Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms. If

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c,$$

where c is a finite number and $c > 0$, then either both series converge or both diverge.

Problem 4.7. Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} \frac{1}{n!}$$

Proof. We use the comparison test. For $n \geq 1$, we see

$$2^{n-1} \leq n! \Rightarrow \frac{1}{n!} \leq \frac{1}{2^{n-1}}$$

Thus

$$\sum_{n=1}^{\infty} \frac{1}{n} \leq \sum_{n=1}^{\infty} \frac{1}{2^{n-1}} < \infty$$

as it is a geometric series with ratio $r = \frac{1}{2} < 1$, hence convergent. □

Problem 4.8. Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$$

Proof. We use the limit comparison test.

$$\lim_{n\to\infty} \frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}} = 1$$

Thus applying the limit commparison theorem, we know the series

$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right) \text{ and } \sum_{n=1}^{\infty} \frac{1}{n}$$

either both converge or both diverge. Since $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, we know

$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right) \text{ diverges}$$

□

Proposition 4.4. If the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n = b_1 - b_2 + b_3 - b_4 + b_5 - b_6 + \cdots, \qquad b_n > 0$$

satisfies

(i) $b_{n+1} \leq b_n$ for all n,

(ii) $\lim_{n\to\infty} b_n = 0$,

then the series is convergent.

Problem 4.9. Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} (-1)^{n+1} n e^{-n}$$

Proof. We'd like to use the alternating series test. We need to check

$$b_n = \frac{n}{e^n}$$

satisfy

1. $\lim_{n\to\infty} b_n = 0$.
2. $b_{n+1} \leq b_n$ for all n.

For the first check:

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{n}{e^n} = \lim_{n\to\infty} \frac{1}{e^n} = 0$$

by L'Hospital. For the second check,

$$\frac{b_{n+1}}{b_n} = \frac{(n+1)e^n}{e^{n+1}n} = \frac{n+1}{n} \cdot \frac{1}{e} \leq 1$$

for all n. Thus the alternating series test applies, hence the series is convergent. □

Week 9 (11/2-11/6)

Definition 4.2. A power series centered at $x = 0$ is an infinite series $\sum_{n=0}^{\infty} c_n x^n$, where c_n are constant coefficients. A power series centered at $x = a$ is of the form $\sum_{n=0}^{\infty} c_n (x-a)^n$.

You can think of the power series as a function who is only defined in some regions. (For a function to make sense, we need to assign a finite value to every point of x, but that is not necessarily the case for power series, hence limited domain).

Example 4.1. If $|x| < 1$, the geometric series

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

where the LHS is a power series centered at 0, with $c_n = 1$.

$$f(x) = 1 + x + x^2 + x^3 + \dots$$

Why do we care about power series? Next topic: Taylor series. Sometimes we can write a complicated function with a power series. One nice thing about power series is that we can differentiate/integrate more easily!

Example 4.2. You will learn to compute this later:

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$$

for all $x \in \mathbb{R}$.

Example 4.3.

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^6}{6!}$$

for all $x \in \mathbb{R}$. For x close to 0, we know intuitively why

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

Are there functions that don't have a power series expansion?

Example 4.4. A smooth function that doesn't have a power series expansion:

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}}, x \neq 0 \\ 0, x = 0 \end{cases}$$

Proposition 4.5. Given a power series $\sum_{n=0}^{\infty} c_n (x-a)^n$, one of three following must happen:

1. The series converges for all $x \in \mathbb{R}$.
2. The series only converges for $x = a$.
3. There exists an $R > 0$ such that whenever $|x - a| < R$, the series converges; when $|x - a| > R$, the series diverges; when $|x - a| = R$, we don't know.

This radius of convergence is closely related to series convergence and all the tests introduced before such as ratio and root tests.

Example 4.5.

$$\sum_{n=0}^{\infty} n!x^n$$

Problem 4.10. Find the radius of convergence for the following power series

$$\sum_{n=1}^{\infty} n^n x^n$$

Proof.

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}(n+1)\left(1+\frac{1}{n}\right)^n|x| = \infty$$

for any $x \neq 0$. Hence the radius of convergence is 0. □

Problem 4.11. Find for which x the following power series converge.

$$\sum_{n=1}^{\infty} \frac{x^{2n}}{n!}$$

Proof. For $x \neq 0$,

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{x^2}{n+1} = 0$$

for any fixed $x \neq 0$, this converges. It also converges at $x = 0$. Thus the power series converge for all $x \in \mathbb{R}$. □

Problem 4.12. If k is a positive integer, find the radius of convergence of the series

$$\sum_{n=0}^{\infty} \frac{(n!)^k}{(kn)!}x^n.$$

Proof.

$$\lim_{n\to\infty}\frac{(n+1)!^k(kn)!}{(kn+k)!(n!)^k}|x| = \lim_{n\to\infty}\frac{(n+1)^k}{(kn+1)\dots(kn+k)}|x| = \frac{1}{k^k}|x|$$

which means the radius of convergence is k^k.

□